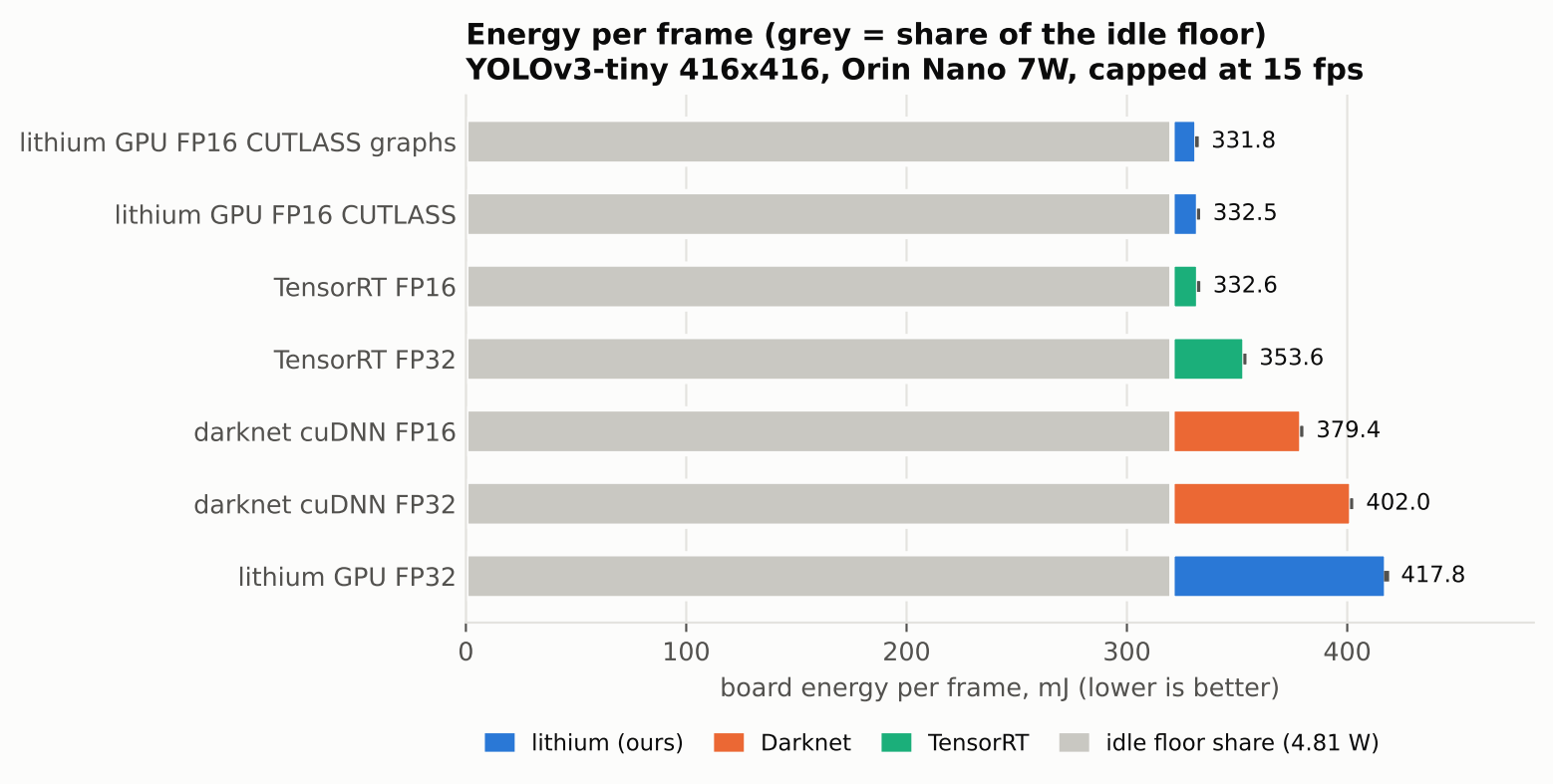
Energy per frame (grey = share of the idle floor)
YOLOv3-tiny 416x416, Orin Nano 7W, capped at 15 fps
lithium GPU FP16 CUTLASS graphs
lithium GPU FP16 CUTLASS
TensorRT FP16
TensorRT FP32
darknet cuDNN FP16
darknet cuDNN FP32
lithium GPU FP32
331.8
332.5
332.6
353.6
379.4
402.0
417.8
0
100
200
300
400
board energy per frame, mJ (lower is better)
lithium (ours)
Darknet
TensorRT
idle floor share (4.81 W)